

Name: Abul Hasan Shaik

Course: CSC474/574 Computer Networks

Assignment: Programming Assignment 1 – Cloud/Network File System

Semester: Spring 2025

Project Git Repository:

https://github.com/AbulHasanShaik/CSC474_574_ProAssgn_AbulHasanShaik

1. How to Compile

Open PowerShell or Command Prompt in the project directory and run:

```
g++ server.cpp file_system.cpp -o server.exe -lws2_32 -std=c++17
```

```
g++ client.cpp -o client.exe -lws2_32 -std=c++17
```

These commands will generate server.exe and client.exe using MinGW on Windows.

2. How to Run

Step 1: Start the server

```
./server.exe
```

Output:

```
Server listening on port 8080...
```

Step 2: Start the client

Open a second terminal and run:

```
./client.exe
```

You'll see:

```
Connected to server. Type commands (mkdir <dir>, ls <dir>, write <file> <data>, read <file>,  
move <src> <dest>, exit):
```

3. Supported Commands

Command Example	Description
<code>mkdir testdir</code>	Creates a directory named testdir
<code>write testdir/file1.txt HelloWorld</code>	Writes HelloWorld to file1.txt
<code>read testdir/file1.txt</code>	Reads content from file1.txt
<code>ls testdir</code>	Lists all files inside testdir
<code>move testdir/file1.txt testdir/file2.txt</code>	Renames or moves file1.txt to file2.txt
<code>exit</code>	Gracefully closes the connection

```

PS C:\Users\shaik\Computer Networks> g++ server.cpp file_system.cpp -o server.exe -lws2_32 -std=c++17
PS C:\Users\shaik\Computer Networks> g++ client.cpp -o client.exe -lws2_32 -std=c++17
PS C:\Users\shaik\Computer Networks> ./server.exe
Server listening on port 8080...
Client connected
Client disconnected
[]

PS C:\Users\shaik\Computer Networks> ./client.exe
Connected to server. Type commands (mkdir <dir>, ls <dir>, write <file> <data>, read <file>, move <src> <dest>, exit):
mkdir testdir
Server: Directory created

write testdir/file1.txt HelloWorld
Server: File written

ls testdir
Server: file1.txt

read testdir/file1.txt
Server: HelloWorld
move testdir/file1.txt testdir/file2.txt
Server: Move successful

ls testdir
Server: file2.txt

read testdir/file2.txt
Server: HelloWorld
exit
PS C:\Users\shaik\Computer Networks>

```

4. Features Implemented

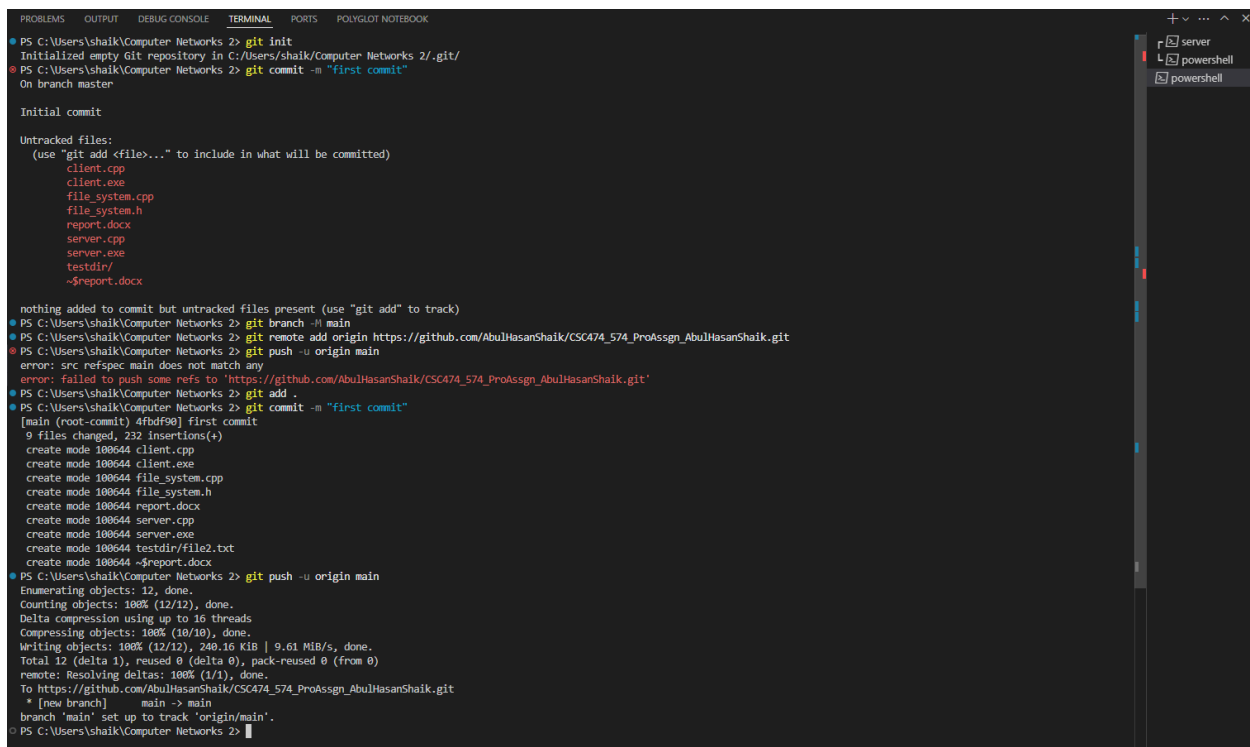
- TCP socket communication using Winsock on Windows
- Multi-client support via multithreaded server
- File system operations using `std::filesystem`:
 - Create directories

- Write to and read from files
- List files in a directory
- Move or rename files
- Clean server-client protocol with newline-based parsing
- Proper error handling (e.g., file not found, invalid input)

5. Project Management Tool Used

Used **Git** locally to track file versions and manage project changes. Commands used:

- git init
- git add, git commit



```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POLYGLOT NOTEBOOK
PS C:/Users/shaik/Computer Networks 2> git init
Initialized empty Git repository in C:/Users/shaik/Computer Networks 2/.git/
PS C:/Users/shaik/Computer Networks 2> git commit -m "first commit"
On branch master

Initial commit

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        client.cpp
        client.exe
        file_system.cpp
        file_system.h
        report.docx
        server.cpp
        server.exe
        testdir/
        ~$report.docx

nothing added to commit but untracked files present (use "git add" to track)
PS C:/Users/shaik/Computer Networks 2> git branch -M main
PS C:/Users/shaik/Computer Networks 2> git remote add origin https://github.com/AbulHasanShaik/CSC474_574_ProAssgn_AbulHasanShaik.git
PS C:/Users/shaik/Computer Networks 2> git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/AbulHasanShaik/CSC474_574_ProAssgn_AbulHasanShaik.git'
PS C:/Users/shaik/Computer Networks 2> git add .
PS C:/Users/shaik/Computer Networks 2> git commit -m "first commit"
[main (root-commit) 4fbd90] first commit
 9 files changed, 232 insertions(+)
 create mode 100644 client.cpp
 create mode 100644 client.exe
 create mode 100644 file_system.cpp
 create mode 100644 file_system.h
 create mode 100644 report.docx
 create mode 100644 server.cpp
 create mode 100644 server.exe
 create mode 100644 testdir/file2.txt
 create mode 100644 ~$report.docx
PS C:/Users/shaik/Computer Networks 2> git push -u origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 16 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (12/12), 240.16 KiB | 9.61 MiB/s, done.
Total 12 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/AbulHasanShaik/CSC474_574_ProAssgn_AbulHasanShaik.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
PS C:/Users/shaik/Computer Networks 2>

```

6. Testing Summary

All commands were tested in sequence:

mkdir testdir

write testdir/file1.txt HelloWorld

ls testdir

read testdir/file1.txt

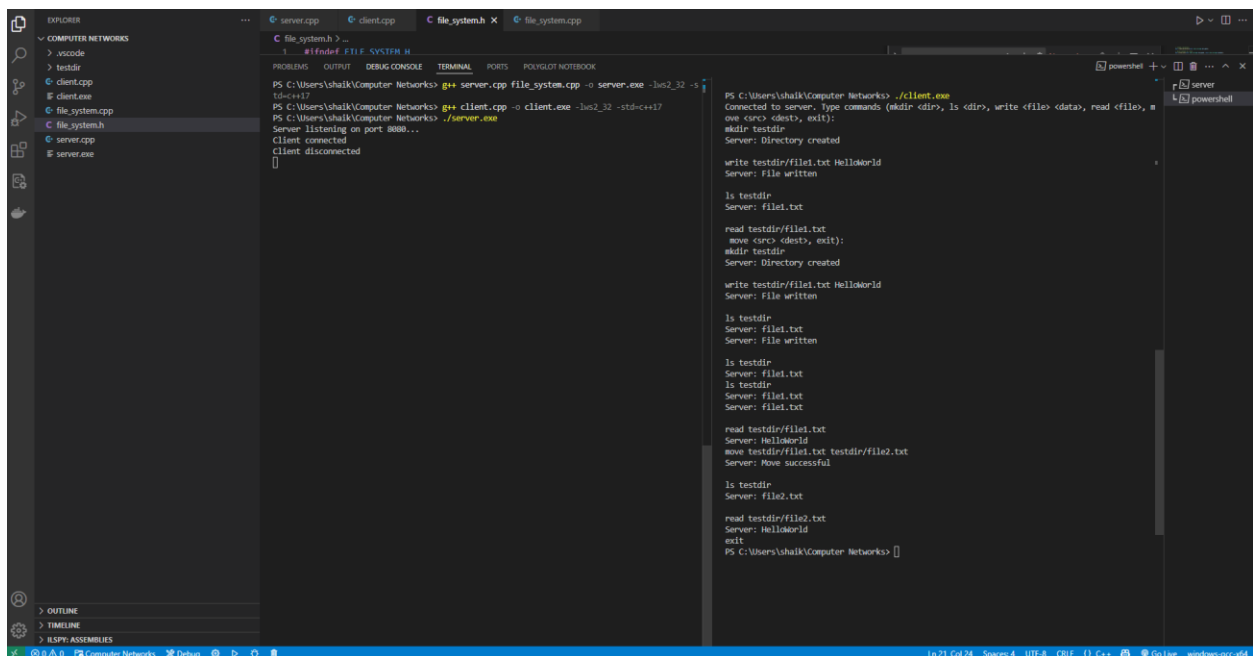
move testdir/file1.txt testdir/file2.txt

ls testdir

read testdir/file2.txt

exit

All commands worked as expected with correct server responses.



```
PS C:\Users\shaik\Computer Networks> g++ server.cpp file_system.cpp -o server.exe -std=c++17
PS C:\Users\shaik\Computer Networks> g++ client.cpp -o client.exe -std=c++17
Server: Listening on port 8080...
Client connected
Client disconnected

PS C:\Users\shaik\Computer Networks> .\client.exe
Connected to server. Type commands (mkdir <dir>, ls <dir>, write <file> <data>, read <file>, move <src> <dest>, exit):
mkdir testdir
Server: Directory created

write testdir/file1.txt HelloWorld
Server: File written

ls testdir
Server: file1.txt

read testdir/file1.txt
move <src> <dest>, exit):
mkdir testdir
Server: Directory created

write testdir/file1.txt HelloWorld
Server: File written

ls testdir
Server: file1.txt
Server: File written

ls testdir
Server: file1.txt
ls testdir
Server: file1.txt
Server: file1.txt
Server: file1.txt

read testdir/file1.txt
Server: HelloWorld
move testdir/file1.txt testdir/file2.txt
Server: Move successful

ls testdir
Server: file2.txt

read testdir/file2.txt
Server: HelloWorld
exit
PS C:\Users\shaik\Computer Networks>
```

7. Files Included in Submission

CSC474_574_ProAssgn_AbulHasanShaik/

└─ client.cpp

└─ server.cpp

└─ file_system.cpp

└─ file_system.h

└─ report.pdf/doc

8. Known Limitations

- Does not support subdirectory recursion (e.g., nested ls)
 - File uploads/downloads via binary stream not implemented (only text supported)
 - Basic plain-text command interface (no encryption/authentication)
-

9. References

- <https://www.geeksforgeeks.org/network-file-system-nfs/>
- <https://people.cs.rutgers.edu/~pxk/416/notes/18-nfs.html>
- <https://github.com/dmonaldo/simple-network-file-system>